

ILAM, Iran, Islamic Republic of

WMO: 407800

Lat: 33.5884N

Lon: 46.3972E

Elev: 1337

StdP: 86.26

Time Zone: 3.50 (IRN)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.3	-2.8	-14.8	1.2	6.7	-12.2	1.5	7.1	9.0	7.5	7.9	8.1	1.4	100	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	39.0	16.9	38.0	16.8	36.9	16.5	18.1	36.7	17.5	35.9	16.9	35.0	4.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.2	10.4	16.9	11.2	9.7	16.8	10.2	9.1	17.2	56.7	36.9	54.7	36.1	52.4	35.0	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.2	6.3	-7.8	40.5	2.6	1.2	-9.7	41.4	-11.2	42.1	-12.7	42.8	-14.6	43.7		
			-9.0	19.6	2.6	1.3	-10.9	20.5	-12.5	21.3	-13.9	22.1	-15.8	23.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	4.7	6.3	10.3	14.7	20.6	26.3	29.4	29.2	24.9	19.1	11.0	7.1	(6)
(7)		DBStd	9.26	3.23	3.61	3.80	3.28	3.14	2.36	2.08	1.87	2.73	3.24	3.38	3.06	(7)
(8)		HDD10.0	447	166	111	40	3	0	0	0	0	0	0	28	99	(8)
(9)		HDD18.3	1739	424	338	249	116	14	0	0	0	0	31	219	347	(9)
(10)		CDD10.0	3013	1	7	51	144	328	489	602	594	446	282	59	10	(10)
(11)		CDD18.3	1263	0	0	2	8	84	239	343	336	196	55	0	0	(11)
(12)		CDH23.3	17558	0	0	16	99	1046	3360	5101	4853	2487	593	3	1	(12)
(13)		CDH26.7	10007	0	0	1	19	364	1914	3201	3040	1315	153	0	0	(13)
(14)	Wind	WSAvg	2.7	2.4	2.5	2.8	2.7	2.7	3.3	3.2	3.1	2.8	2.5	2.2	2.2	(14)
(15)	Precipitation	PrecAvg	490	105	70	75	55	13	0	0	1	1	26	69	70	(15)
(16)		PrecMax	809	180	194	192	127	63	2	7	17	17	191	354	168	(16)
(17)		PrecMin	275	19	12	0	4	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	127	42	43	57	37	17	0	1	3	3	39	70	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.9	18.6	25.2	29.2	33.8	38.2	40.2	39.9	37.2	31.8	23.2	20.2	(19)
(20)		MCWB	6.3	8.8	11.0	13.5	15.3	16.8	17.4	17.3	16.9	14.6	10.5	8.3	(20)	
(21)		2%	DB	14.0	16.2	22.2	26.0	32.0	37.0	39.2	38.9	35.9	29.9	21.6	17.1	(21)
(22)		MCWB	5.7	7.1	9.8	12.8	14.5	16.2	17.1	16.8	16.0	13.5	9.9	7.3	(22)	
(23)		5%	DB	12.1	14.8	19.8	23.9	30.2	35.8	38.1	38.0	34.3	28.2	19.8	14.9	(23)
(24)		MCWB	5.3	6.6	9.0	12.0	13.9	15.8	16.9	16.7	15.3	12.8	9.5	6.8	(24)	
(25)		10%	DB	10.8	12.9	17.9	22.0	28.8	34.5	37.0	36.9	33.0	26.9	17.8	12.8	(25)
(26)		MCWB	5.0	6.2	8.3	11.0	13.7	15.3	16.6	16.5	14.7	12.3	9.3	6.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	10.6	12.1	14.9	16.5	17.8	19.2	18.9	18.3	15.7	13.6	11.2	(27)
(28)		MCDWB	11.5	14.8	19.3	23.8	28.5	36.3	38.4	37.6	34.4	26.9	17.7	13.8	(28)	
(29)		2%	WB	7.8	9.2	11.1	13.9	15.6	17.1	18.2	18.1	16.7	14.6	12.3	9.3	(29)
(30)		MCDWB	10.3	13.1	18.6	22.5	28.2	35.1	37.2	36.7	33.6	25.0	16.2	13.4	(30)	
(31)		5%	WB	6.7	7.8	10.3	13.0	15.0	16.4	17.5	17.5	15.9	13.9	11.5	8.2	(31)
(32)		MCDWB	9.9	12.8	17.5	20.7	27.3	34.3	36.2	35.9	32.8	24.0	15.8	12.6	(32)	
(33)		10%	WB	5.6	6.9	9.4	12.2	14.4	15.7	16.9	16.9	15.1	13.3	10.5	7.4	(33)
(34)		MCDWB	9.6	11.7	16.1	19.5	26.2	33.3	35.4	35.1	31.7	23.5	15.5	11.5	(34)	
(35)	Mean Daily Temperature Range		MDBR	10.0	10.3	11.3	11.9	13.4	15.0	14.6	14.9	14.9	12.9	11.2	10.6	(35)
(36)		5% DB	MCDBR	12.1	12.7	13.6	14.1	15.2	16.3	15.3	15.4	15.4	14.4	13.8	13.7	(36)
(37)			MCWBR	6.9	6.7	6.5	6.3	6.2	6.7	6.1	6.1	6.6	6.4	6.9	7.4	(37)
(38)			MCDBR	9.3	10.5	10.8	11.7	13.3	15.5	14.8	14.9	14.7	11.7	9.2	9.9	(38)
(39)		5% WB	MCWBR	6.2	6.4	5.9	5.9	5.7	6.7	6.3	6.4	6.7	5.7	5.3	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.373	0.427	0.511	0.501	0.440	0.488	0.445	0.392	0.426	0.363	0.339	(40)	
(41)		taud	2.238	2.114	1.955	1.786	1.827	1.969	1.849	1.961	2.094	1.997	2.171	2.233	(41)	
(42)		Ebn at Noon	863	862	841	787	800	848	805	834	865	791	817	832	(42)	
(43)		Edh at Noon	109	138	176	218	212	183	205	180	151	153	116	104	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.87	3.77	5.13	5.96	7.08	8.34	7.89	7.36	6.36	4.58	3.35	2.71	(44)	
(45)		RadStd	0.18	0.28	0.35	0.47	0.40	0.24	0.23	0.22	0.22	0.35	0.34	0.23	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (3 neighbors)	N/A	N/A	-3.48	+0.56	-1.68	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page